way that air carries sound waves, a Higgs ocean has nothing to do with the motion of light; it does not affect light's speed in any way, and so experiments from the turn of the twentieth century that ruled out the aether by studying light's motion have no bearing on the Higgs ocean.

Moreover, since the Higgs ocean has no effect on anything moving with constant velocity, it does not pick out one observational vantage point as somehow being special, as the aether did. Instead, even with a Higgs ocean, all constant velocity observers remain on a completely equal footing, and hence a Higgs ocean does not conflict with special relativity. Of course, these observations do not prove that Higgs fields exist; instead, they show that despite certain similarities to the aether, Higgs fields are not in conflict with any theory or experiment.

If there is an ocean of Higgs field, though, it should yield other consequences that will be experimentally testable within the next few years. As a primary example, just as electromagnetic fields are composed of photons, Higgs fields are composed of particles that, not surprisingly, are called *Higgs particles*. Theoretical calculations have shown that if there is a Higgs ocean permeating space, Higgs particles should be among the debris from the high-energy collisions that will take place at the Large Hadron Collider, a giant atom smasher now under construction at Centre Européène pour la Recherche Nuclaire (CERN) in Geneva, Switzerland, and slated to come online in 2007. Roughly speaking, enormously energetic head-on collisions between protons should be able to knock a Higgs particle out of the Higgs ocean somewhat as energetic underwater collisions can knock H_2O molecules out of the Atlantic. In due course, these experiments should allow us to determine whether this modern form of the aether exists or whether it will go the way of its earlier incarnation. This is a critical question to settle because, as we have seen, condensing Higgs fields play a deep and pivotal role in our current formulation of fundamental physics.

If the Higgs ocean is not found, it will require major rethinking of a theoretical framework that has been in place for more than thirty years. But if it is found, the event will be a triumph for theoretical physics: it will confirm the power of symmetry to correctly shape our mathematical reasoning as we venture forth into the unknown. Beyond this, confirmation of the Higgs ocean's existence would also do two more